Cayley–Hamilton — Reduce Every Power

READ IN ORDER: top to bottom in the left column, then top to bottom in the right column; then next sheet.

LESSON 01 / 03

Characteristic polynomial

For square A, define $p(\lambda) = \det(\lambda I - A)$. Eigenvalues are roots because $(A - \lambda I)v = 0$ has a nonzero vector exactly when determinant vanishes. For 2×2:

$$p(\lambda) = \lambda^2 - (\operatorname{tr} A)\lambda + \det A.$$

KEY RESULT

Cayley–Hamilton theorem

Every square matrix satisfies its characteristic polynomial: $p(A) = 0$. For order 2, $A^2 - (\operatorname{tr} A)A + (\det A)I = 0$. Therefore every higher power reduces to a linear combination of I and A.

WORKED EXAMPLE

Power reduction

For $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, trace 5, determinant -2, so $A^2 - 5A - 2I = 0$. Then $A^2 = 5A + 2I$ and $A^3 = 27A + 10I$.

Eigenvalue invariants and similarity

$B = P^{-1}AP$ is a change of basis (similarity), when P invertible. Then $\det(\lambda I - B) = \det(\lambda I - A)$; hence trace and determinant, and the eigenvalue multiset, are unchanged. Distinct eigenvalues imply diagonalizability over a field containing all eigenvalues (for real matrices, at least over $\mathbb{C}$); repeated eigenvalues alone do not.

COMMON TRAP

Avoid an invalid substitution

Cayley–Hamilton lets us reduce powers of one fixed square matrix. It does not permit replacing a scalar variable by a matrix in an unrelated identity unless products commute and the polynomial identity applies.

LESSON 02 / 03

Hadamard's volume bound

For columns $c_1, \ldots, c_n$ of A, $|\det A| \le \prod_j \|c_j\|$. Volume cannot exceed the rectangular box with those edge lengths; equality holds exactly when nonzero columns are pairwise orthogonal (zero-column case gives both sides zero).

SHORT DERIVATION

Two-dimensional identity

For columns $(a, c)^T, (b, d)^T$, $(ad - bc)^2 = (a^2 + c^2)(b^2 + d^2) - (ab + cd)^2$. Thus $|ad - bc|$ is at most the product of their lengths; equality iff dot product is zero.

Rank-one determinant lemma

For invertible A and compatible column vectors u, v:

$$\det(A + uv^T) = \det A(1 + v^T A^{-1} u).$$

A high-dimensional determinant update reduces to one scalar. Equivalent core identity: $\det(I + XY) = \det(I + YX)$ for compatible rectangular X, Y.

WORKED EXAMPLE

Check one update

$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, $u = (1, 2)^T$, $v = (3, 4)^T$.

Directly $\det(A + uv^T) = -6$; the lemma gives $(-2)(1 + 2) = -6$.

Vandermonde determinant

$$\det \begin{pmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ 1 & x_3 & x_3^2 \end{pmatrix} = (x_2 - x_1)(x_3 - x_1)(x_3 - x_2).$$

It vanishes iff two nodes coincide, exactly when quadratic interpolation loses uniqueness. General form: $\prod_{i<j}(x_j - x_i)$.

COMMON TRAP

State the needed hypotheses

The matrix determinant lemma uses A^{-1}, so A must be invertible. The Vandermonde product depends on the stated column order $1, x, x^2$ and row order x_1, x_2, x_3; reordering nodes changes sign accordingly.

LESSON 03 / 03

Trace: cyclic, not commutative

For square matrices A, B of the same order, $\operatorname{tr}(AB) = \operatorname{tr}(BA)$ since both equal $\sum_{i,j} a_{ij} b_{ji}$. This does **not** say $AB = BA$. More generally $\operatorname{tr}(ABC) = \operatorname{tr}(BCA)$ when dimensions permit cyclic products.

Rank–nullity

For linear map $A : \mathbb{R}^n \to \mathbb{R}^m$, $\operatorname{rank} A + \operatorname{nullity} A = n$. Row reduction exposes pivot variables and free variables; the null space is the solution space of $Ax = 0$.

WORKED EXAMPLE

Dimension count

If a 3×4 matrix has rank 2, then nullity is $4 - 2 = 2$; the homogeneous system has a two-dimensional family of solutions.

Block determinants

For square diagonal blocks A, D:

$$\det\begin{pmatrix} A & B \\ 0 & D \end{pmatrix} = \det A \det D.$$

If A is invertible (and block sizes compatible), Schur complement gives

$$\det\begin{pmatrix} A & B \\ C & D \end{pmatrix} = \det A \det(D - CA^{-1}B).$$

INTUITION

Eliminate the lower-left block

Multiply on the left by the block triangular matrix $\begin{pmatrix} I & 0 \\ -CA^{-1} & I \end{pmatrix}$, whose determinant is 1.

The product has lower-left block zero and lower-right block $D - CA^{-1}B$.

WORKED EXAMPLE

Schur complement in one calculation

Let $A = I_2$, $B = (1,2)^T$, $C = (3,4)$, $D = (10)$. Then the Schur complement is $10 - (3,4)(1,2)^T = 10 - 11 = -1$, so the block determinant is -1. Directly,

$$\det\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 3 & 4 & 10 \end{pmatrix} = -1.$$

Boundary reminder

Eigenvalues, Cayley–Hamilton, rank–nullity, Hadamard, Vandermonde and Schur complements are useful Olympiad extensions; they are not being represented as explicit JEE syllabus requirements. Matrix algebra can still arise inside a JEE question when supplied or derivable.